# SymPy Bug Card

## Bug 48: Core algebraic identities erase incomplete Piecewise undefined branches

**Task.** Evaluate core identities built from an incomplete `Piecewise` on its implicit undefined branch.

$$p = \mathrm{Piecewise}((1, x > 0)), \qquad x = -1$$

**SymPy's answer.**

$$p(-1) = \mathrm{NaN}, \qquad (p \cdot 0)(-1) = 0, \qquad (p - p)(-1) = 0, \qquad \mathrm{Eq}(p, p)(-1) = \mathrm{True}.$$

**Correct answer.**

$$(p \cdot 0)(-1) = \mathrm{NaN}, \qquad (p - p)(-1) = \mathrm{NaN}, \qquad \mathrm{Eq}(p, p)(-1) \neq \mathrm{True}.$$

**Explanation.** The expression $p$ has no branch for $x \leq 0$, and SymPy already evaluates $p(-1)$ as NaN. The finite-value identities $z \cdot 0 = 0$, $z - z = 0$, and $z = z$ are not valid at a point where $z$ is undefined, so these simplifications incorrectly extend the domain.

**Likely root cause.** The diagnosis points to generic core simplification: `sympy/core/mul.py` collapses multiplication and cancellation without checking for incomplete `Piecewise` operands, while `sympy/core/relational.py` turns structurally identical operands into `True`. Deduplication frames this as a new instance of a known failure mode.

**Artifact (GitHub).** `bug-report/bug-048-core-algebraic-identities-erase-incomplete-piecewise-undefin/`